Math 142, F19 **Name:** Answer
Quiz 3

<u>Instructions:</u> **You must show all of your work to receive credits for correct answers.**

Problem 1. (10 points) Evaluate the integrals

(a) $\int \sin^5\theta\, d\theta$. (*Hint: use the substitution* $u = \cos\theta$)

$u = \cos\theta$, $du = (-\sin\theta)d\theta$

$$\Rightarrow \int \sin^5\theta\, d\theta = \int \sin^4\theta(\sin\theta\, d\theta) = -\int (\underbrace{\sin^2\theta}_{=1-\cos^2\theta})^2(-\sin\theta\, d\theta)$$

$$= -\int (1-u^2)^2 du = -\int 1-2u^2+u^4 du$$

$$= -\left(u - \frac{2}{3}u^3 + \frac{1}{5}u^5\right) + C$$

$$= -\cos\theta + \frac{2}{3}\cos^3\theta - \frac{1}{5}\cos^5\theta + C$$

(b) $\int \tan x \sec^4 x\, dx$. (*Hint: use the substitution* $u = \tan x$)

$u = \tan x$, $du = \sec^2 x\, dx$

$$\Rightarrow \int \tan x \sec^4 x\, dx = \int \tan x \sec^2 x(\sec^2 x\, dx)$$

$$= \int \tan x(\tan^2 x + 1)\sec^2 x\, dx$$

$$= \int u(u^2+1)\, du$$

$$= \int u^3 + u\, du = \frac{u^4}{4} + \frac{u^2}{2} + C$$

$$= \frac{1}{4}\tan^4 x + \frac{1}{2}\tan^2 x + C$$

(c) $\int_0^{\pi/2} \sqrt{\frac{1+\cos x}{2}}\, dx$. (*Hint: use a half-angle identity*)

$$= \int_0^{\pi/2} \sqrt{\cos^2\left(\frac{x}{2}\right)}\, dx = \int_0^{\pi/2} \cos\left(\frac{x}{2}\right) dx$$

$$= 2\sin\left(\frac{x}{2}\right)\Big|_0^{\pi/2} = 2\left(\sin\frac{\pi}{4} - 0\right) = 2\cdot\frac{\sqrt{2}}{2} = \sqrt{2}$$